

Write a function which will accept this string and return a revised string in below format, "CAF SoftSol India Pvt Ltd."

```
def normalize_company_name(company_name):  
    """  
    'CAF SoftSol India Pvt Ltd.'  
    """  
  
    # Handle empty or null values gracefully  
    if not company_name or not isinstance(company_name, str):  
        return None  
  
    name = company_name.strip().lower()  
  
    # Known CAF variations  
    caf_variations = [  
        "caf softsol",  
        "caf solution",  
        "caf softsolution",  
        "caf softsolution india pvt limited",  
        "caf softsol india"  
    ]  
  
    if any(variant in name for variant in caf_variations):  
        return "CAF SoftSol India Pvt Ltd."  
  
    # Return formatted value for unknown companies  
    return company_name.title()
```

2. Write testcases for your program. Ex: If column doesn't have an entry then it should be handled gracefully. TESTCASES

```
import unittest  
  
from company_name_normalizer import normalize_company_name
```

```
class TestCompanyNameNormalizer(unittest.TestCase):

    def test_softsol(self):
        self.assertEqual(
            normalize_company_name("CAF softsol"),
            "CAF SoftSol India Pvt Ltd."
        )

    def test_solution(self):
        self.assertEqual(
            normalize_company_name("CAF solution"),
            "CAF SoftSol India Pvt Ltd."
        )

    def test_full_name(self):
        self.assertEqual(
            normalize_company_name("CAF softSolution India Pvt Limited"),
            "CAF SoftSol India Pvt Ltd."
        )

    def test_empty_value(self):
        self.assertIsNone(normalize_company_name(""))

    def test_none_value(self):
        self.assertIsNone(normalize_company_name(None))

    def test_unknown_company(self):
        self.assertEqual(
            normalize_company_name("ABC Technologies"),
            "Abc Technologies"
```

)

if __name__ == "__main__":

unittest.main()